

Majid Daliri

370 Jay St, Brooklyn, NY
11th Floor, Desk A1119

Cell: +1-646-750-4667
daliri.majid@nyu.edu
[majid-daliri.github.io](https://github.com/majid-daliri)

Education	New York University, New York, USA Ph.D. in Computer Science (in progress) Student in the Theoretical Computer Science group Advised by Prof. Christopher Musco	2022 - present
	New York University, New York, USA M.S. in Computer Science	2022 - 2024
	University of Tehran, Tehran, Iran B.S. in Computer Engineering (GPA: 3.97/4.0)	2017 - 2022
Publications	(1) Coupling without Communication and Drafter-Invariant Speculative Decoding (Arxiv) Majid Daliri, Christopher Musco, Ananda Theertha Suresh (2) Unlocking the Theory Behind Scaling 1-Bit Neural Networks (CPAL) 2025 Majid Daliri, Zhao Song, Chiwun Yang (3) QJL: 1-Bit Quantized JL transform for KV Cache Quantization (AAAI) 2025 Amir Zandieh, Majid Daliri, Insu Han (4) Matrix Product Sketching via Coordinated Sampling (ICLR) 2025 Majid Daliri, Juliana Freire, Danrong Li, Christopher Musco (5) Sampling Methods for Inner Product Sketching (VLDB) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang (6) Simple Analysis of Priority Sampling (SOSA) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang (7) KDEformer: Accelerating Transformers via Kernel Density Estimation (ICML) 2023 Amir Zandieh, Insu Han*, Majid Daliri*, Amin Karbasi (* equal contribution) (8) Weighted Minwise Hashing Beats Linear Sketching for Inner Product Estimation (PODS) 2023 Aline Bessa, Majid Daliri, Juliana Freire, Cameron Musco, Christopher Musco, Aécio Santos, Haoxiang Zhang (9) Efficient Approximations for Cache-conscious Data Placement (PLDI) 2022 Ali Ahmadi, Majid Daliri, Amir Kafshdar Goharshady, Andreas Pavlogiannis (10) A 10-Approximation of the $\frac{\pi}{2}$-MST (STACS) 2022 Ahmad Biniaz, Majid Daliri, AmirHossein Moradpour	
Research Internship	Machine Learning Intern, Max Planck Institute, Advised by Dr. Amir Zandieh	2021-2022
	- Implemented Fast Attention in Transformers, optimizing accuracy and efficiency in sequence modeling tasks. - Designed an efficient GPU-compatible LSH method, boosting performance in attention approximation. - Technical Stack: BigGAN, PyTorch, Transformer architectures, and advanced sequence modeling tools.	
	Research Intern, HKUST, Advised by Prof. Amir Goharshady	2021-2022
	- Made a pioneering theoretical contribution to Cache-conscious Data Placement (CDP), addressing a longstanding challenge in optimizing cache hits. - Implemented and tested various cache management policies and algorithms, outperforming previous heuristics. - Our method emerged as the most effective solution, setting a new standard for cache optimization.	
	Research Intern, University of Salzburg, Advised by Prof. Ana Sokolova	Summer 2022
	- Developed algorithms for Distribution Bisimilarity, focusing on finite bisimulation up to convex hull witness. - Extended research to probabilistic system verification and collaborated on using Quantum Annealers for program verification.	

Awards and Honors	Apple Scholars in AI/ML	2025-2027
	Full graduate fellowship for the 2025–2027 academic years.	
	Travel Grants	2024, 2025
	Awarded travel grants to present papers at: • AAAI Conference on Artificial Intelligence (2025) • ACM-SIAM Symposium on Discrete Algorithms (2024) • ICERM Fusing Theory and Practice of Graph Algorithms (2025)	
	Research Grant, University of Salzburg	Summer 2022
	Awarded a €5,000 grant for a research internship focusing on algorithms for distribution bisimilarity, probabilistic systems verification, and quantum annealing projects.	
	Hong Kong PhD Fellowship Scheme (declined to attend NYU)	2022
	Awarded \$185,000 Ph.D. Fellowship as one of the top 300 students selected across all fields for academic excellence and research potential.	
	ACM ICPC - Regional (University of Tehran)	2019
	Ranked 6 th among more than 100 team all around the Iran.	
	Iranian National Olympiad in Informatics Finalist (IOI, Iran)	2016
	Awarded to 50 students after a year long competition involving 10,000 students.	
Service	• Reviewer for International Conference on Learning Representations (ICLR 2025) • Reviewer for Conference on Neural Information Processing Systems (NeurIPS 2024) • Reviewer for ACL Rolling Review (ACL, EMNLP, NAACL 2024-2025) • Reviewer for International Conference on Machine Learning (ICML 2024-2025) • Reviewer for Royal Society Open • External Reviewer for Canadian Conference on Computational Geometry (CCCG 2023)	
Conference Presentations	• QJL: 1-Bit Quantized JL transform for KV Cache Quantization • Coupling without Communication • Simple Analysis of Priority Sampling • Accelerating Transformers via Kernel Density Estimation • Weighted MinHash for Inner Product Estimation • Efficient Approximations for Cache-conscious Data Placement	Poster, AAAI 2025 Presentation, ICERM 2025 - Presentation, (NYC TCS Day) 2024 - Presentation, (SOSA) 2024 - Poster, (ICML) 2023 - Poster, (PODS) 2023 - Presentation, (PLDI) 2022
Teaching	• Section Leader for NYU CSCI-UA 310 Basic Algorithms • Teaching Assistant NYU CS-GY 6763 Algorithmic Machine Learning • Teaching Assistant University of Tehran Design and Analysis of Algorithms	- Spring 2023 - Fall 2022 - Fall 2020-2021
Work Experience	Site Reliability Engineer at Cafebazaar • Designed, implemented, and maintained both Redis-as-a-Service/PostgreSQL-as-a-Service on a Kubernetes-based cloud. • Achieved consistent performance benchmarks for both services with 100% uptime and a 99.9% response rate. • Technical Stack: Kubernetes, Docker, Sentry, S3, Prometheus	2021 - 2022
Skills	Theoretical Background: Proficient in Machine Learning Theory, Neural Networks, Linear Algebra, and Probability.	
	Technical Skills: Highly skilled in C/C++, CUDA, Go, Python, Bash-Scripting, PHP, JavaScript. Experience with PyTorch, TensorFlow, Django, CSS3, HTML5, and git.	